

SHOYNA, Russian Federation

WMO: 222710

Lat: 67.877N

Lon: 44.174E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.2	-26.5	-31.8	0.2	-29.2	-28.9	0.3	-26.4	18.1	-9.2	15.5	-10.3	2.0	90	0.900

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	20.8	15.9	18.4	14.8	16.5	13.7	16.8	19.7	15.2	17.7	13.9	16.2	5.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.0	18.5	14.0	10.0	16.9	12.7	9.2	15.3	47.4	19.8	42.9	17.6	39.2	16.2	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.3	11.0	DB	-31.1	23.8	3.3	2.5	-33.5	25.6	-35.4	27.1	-37.2	28.5	-39.6	30.3	(4)
(5)			WB	-31.2	18.4	3.1	1.5	-33.5	19.5	-35.3	20.4	-37.1	21.3	-39.4	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-11.5	-12.0	-8.8	-4.2	0.8	7.2	11.2	10.2	7.2	2.3	-3.7	-7.2	(6)	
	DBStd	9.41	7.07	7.73	5.89	4.31	3.86	3.98	3.84	2.76	2.72	3.37	5.14	6.22	(7)	
	HDD10.0	4016	666	616	582	426	285	106	31	32	90	238	410	533	(8)	
	HDD18.3	6927	924	849	841	676	542	336	225	252	333	496	660	792	(9)	
	CDD10.0	134	0	0	0	0	1	21	67	38	7	0	0	0	(10)	
	CDD18.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(11)	
	CDH23.3	17	0	0	0	0	0	1	16	1	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.4	5.6	5.7	5.4	5.0	5.2	5.4	5.0	4.9	5.2	5.5	5.7	5.7	(14)	
(15) Precipitation	PrecAvg	420	28	24	26	22	31	33	36	47	49	56	36	32	(15)	
	PrecMax	592	75	41	70	54	72	64	85	109	97	99	75	65	(16)	
	PrecMin	332	9	5	7	5	9	9	10	15	12	28	9	10	(17)	
	PrecStd	52	13	9	16	13	15	14	18	24	21	17	14	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	0.8	1.2	3.1	16.0	21.5	25.5	21.0	15.6	9.6	5.6	2.1	(19)	
		MCWB	0.2	0.5	0.4	1.7	11.2	15.3	18.4	15.6	12.8	8.6	4.9	1.4	(20)	
	2%	DB	-0.4	0.0	0.5	1.8	11.5	18.7	22.5	18.0	13.3	8.6	4.1	1.1	(21)	
		MCWB	-1.0	-0.3	-0.1	0.9	8.5	14.0	17.3	15.3	11.7	7.6	3.3	0.5	(22)	
	5%	DB	-1.2	-1.1	-0.3	1.2	8.4	16.1	19.9	16.2	12.2	7.6	3.0	0.4	(23)	
		MCWB	-1.7	-1.5	-0.7	0.5	6.4	12.2	15.8	14.0	10.9	6.7	2.2	-0.1	(24)	
	10%	DB	-2.1	-2.4	-1.0	0.7	6.2	13.7	17.7	14.8	11.2	6.8	2.0	-0.2	(25)	
		MCWB	-2.5	-2.8	-1.5	0.1	4.8	10.8	14.6	13.0	10.1	6.0	1.2	-0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.7	0.7	1.9	11.4	16.6	19.6	17.1	13.6	8.9	4.9	1.6	(27)	
		MCDB	0.6	0.7	1.0	2.4	15.3	19.7	23.6	19.1	14.8	9.4	5.3	1.8	(28)	
	2%	WB	-0.8	-0.2	0.0	1.2	8.7	14.2	17.9	15.4	12.1	7.8	3.4	0.6	(29)	
		MCDB	-0.5	-0.1	0.3	1.5	11.4	17.8	22.1	17.3	13.1	8.4	4.0	1.0	(30)	
	5%	WB	-1.6	-1.4	-0.7	0.7	6.6	12.6	16.4	14.2	11.1	6.8	2.4	0.0	(31)	
		MCDB	-1.2	-1.1	-0.3	1.1	8.4	15.9	19.3	15.9	12.1	7.4	2.9	0.3	(32)	
	10%	WB	-2.5	-2.8	-1.5	0.3	4.8	10.9	14.9	13.2	10.2	6.1	1.4	-0.7	(33)	
		MCDB	-2.1	-2.4	-1.1	0.7	6.1	13.6	17.4	14.7	11.1	6.7	1.9	-0.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.4	6.6	5.5	4.7	5.7	6.2	5.0	4.0	3.3	4.5	5.7	(35)	
		MCDBR	5.7	4.2	4.3	4.0	8.9	10.3	10.2	7.8	5.7	3.1	3.2	3.5	(36)	
		MCWBR	5.5	4.4	4.2	3.7	6.5	6.5	6.1	5.4	4.2	2.9	3.0	3.4	(37)	
	5% WB	MCDBR	5.5	4.2	4.3	3.7	8.7	9.5	9.2	7.1	5.0	2.8	3.2	3.6	(38)	
		MCWBR	5.5	4.3	4.3	3.5	6.5	6.7	6.1	5.2	3.9	2.9	3.2	3.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.219	0.264	0.302	0.330	0.339	0.339	0.370	0.348	0.323	0.292	0.363	N/A	(40)	
	taud		1.924	2.035	2.095	2.157	2.361	2.492	2.429	2.479	2.497	2.244	2.560	N/A	(41)	
	Ebn at Noon		85	513	715	810	853	866	818	800	728	493	107	0	(42)	
	Edh at Noon		10	51	87	109	100	91	94	80	61	42	13	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.03	0.41	1.83	3.56	4.65	5.21	4.83	3.19	1.71	0.53	0.08	0.00	(44)	
	RadStd		0.00	0.07	0.20	0.27	0.30	0.57	0.54	0.34	0.18	0.08	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page